

RQ Answers

omitted

2024-04-05

```
glimpse(all_data)
```

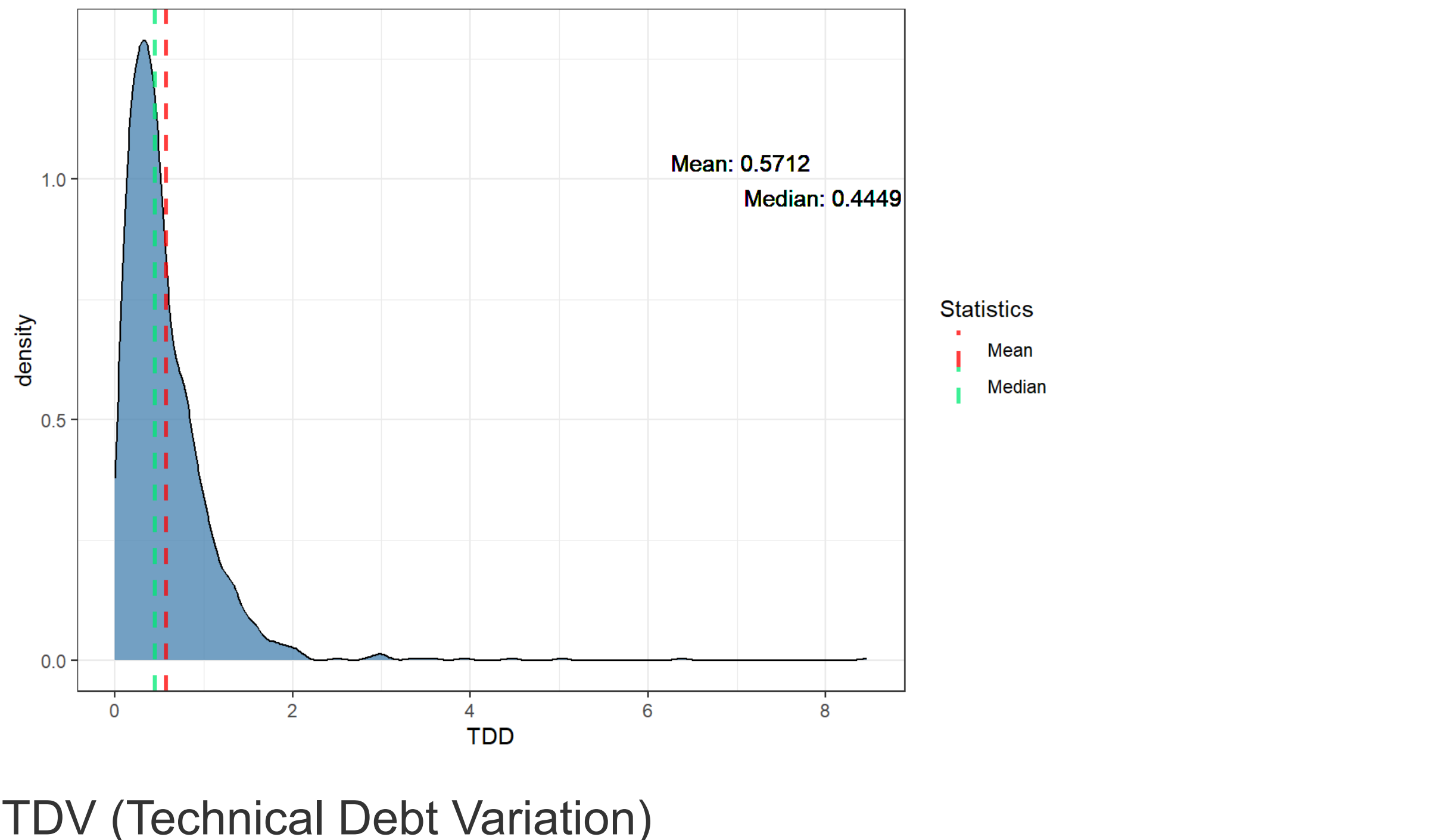
```
## Rows: 71,698
## Columns: 11
## $ repo      <chr> "accumulo", "accumulo", "accumulo", "accumulo", "a...
## $ pr_number <int> 1302, 1302, 1302, 1302, 1302, 1302, 1302, 13...
## $ rule       <chr> "java:S5803", "java:S108", "java:S108", "java:S110...
## $ severity   <chr> "CRITICAL", "MAJOR", "MAJOR", "CRITICAL", "CRITICA...
## $ file       <chr> "server/tserver/src/test/java/org/apache/accumulo/...
## $ type       <chr> "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "CODE_SM...
## $ status     <chr> "OPEN", "CLOSED", "CLOSED", "OPEN", "OPEN", "OPEN"...
## $ debt       <int> 10, 5, 5, 10, 10, 10, 10, 20, 10, 10, 5, 5, 20...
## $ ncloc_affected_file <int> 96, 96, 96, 96, 96, 121, 121, 121, 249, 249, ...
## $ complexity <int> 7, 7, 7, 7, 7, 6, 6, 6, 37, 37, 37, 37, 37...
## $ origin     <chr> "NEW", "PRE-EXISTING", "PRE-EXISTING", "PRE-EXISTI...
```

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd)), : All aesthetics have length 1, but the data has 1152 rows.
## i Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd)), : All aesthetics have length 1, but the data has 1152 rows.
## i Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

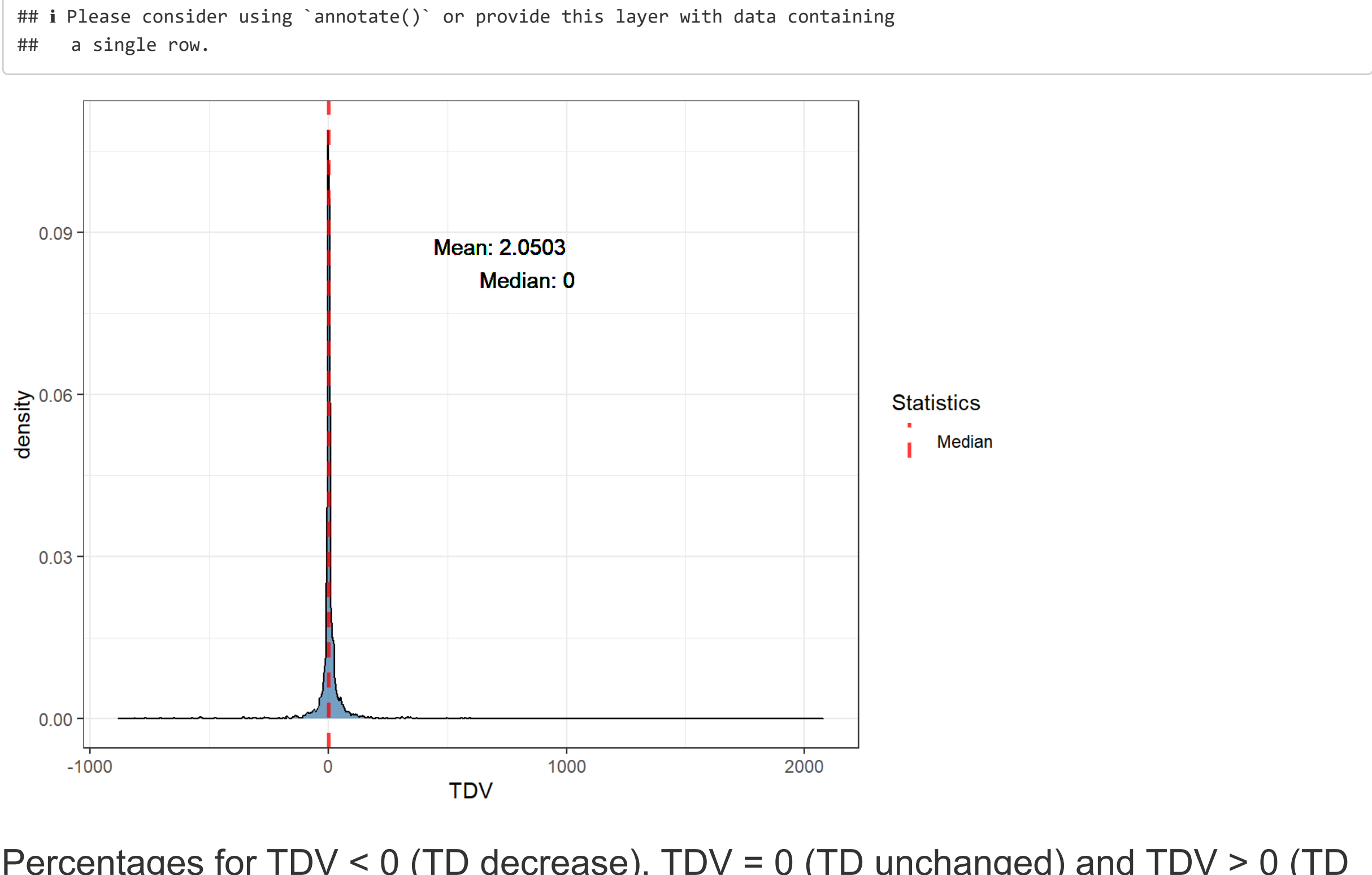


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv)), : All aesthetics have length 1, but the data has 1152 rows.
## i Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv)), : All aesthetics have length 1, but the data has 1152 rows.
## i Please consider using `annotate()` or provide this layer with data containing
##   a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

```
## # A tibble: 12 x 4
##   repo      tdv_lower_than_zero tdv_equal_to_zero tdv_greater_than_zero
##   <chr>          <dbl>          <dbl>          <dbl>
## 1 accumulo          22.0            50.6            27.3
## 2 cayenne          11.1            40.7            48.2
## 3 commons-collections 33.3            50             16.7
## 4 commons-io        28.8            50             21.2
## 5 commons-lang      21.8            63.6            14.6
## 6 helix             13.3            43.4            43.4
## 7 httpcomponents-client 16.7            43.8            39.6
## 8 maven-surefire     25             65             10
## 9 opennlp           26.9            38.5            34.6
## 10 struts            26.7            36.7            37.5
## 11 wicket            12.5            50             37.5
## 12 zookeeper         9.09            54.6            36.4
```

Mean of percentages of TDV (mean of repo percentages)

```
## # A tibble: 1 x 3
##   mean_tdv_lower_than_zero mean_tdv_equal_to_zero mean_tdv_greater_than_zero
##   <dbl>          <dbl>          <dbl>
## 1      20.6            48.9            30.5
```

Percentages of pre-existing TDV by repo

```
## # A tibble: 12 x 3
##   repo      preexisting_tdv_equal_to_zero preexisting_tdv_greater_than_zero
##   <chr>          <dbl>          <dbl>
## 1 accumulo          62.6            37.4
## 2 cayenne          68             32
## 3 commons-collections 50             50
## 4 commons-io        66.7            33.3
## 5 commons-lang      69.1            30.9
## 6 helix             69.8            30.2
## 7 httpcomponents-client 64.4            35.6
## 8 maven-surefire     65             35
## 9 opennlp           64.6            35.4
## 10 struts            56.5            43.5
## 11 wicket            61.5            38.5
## 12 zookeeper        77.3            22.7
## # i abbreviated name: 'preexisting_tdv_greater_than_zero'
```

Mean of percentages of pre-existing TDV (mean of repo percentages)

```
## # A tibble: 1 x 2
##   preexisting_tdv_equal_to_zero mean_preexisting_tdv_greater_than_zero
##   <dbl>          <dbl>
## 1      64.6            35.4
```

Percentages for new TDV by repo

```
## # A tibble: 12 x 3
## # Groups:   repo [12]
##   repo      new_tdv_equal_to_zero new_tdv_greater_than_zero
##   <chr>          <dbl>          <dbl>
## 1 accumulo          83.0            17.0
## 2 cayenne          93.3            6.67
## 3 commons-collections 80             20
## 4 commons-io        88.9            11.1
## 5 commons-lang      84.2            15.8
## 6 helix             86.0            14.0
## 7 httpcomponents-client 95.6            4.35
## 8 maven-surefire     100             0
## 9 opennlp           96.2            3.85
## 10 struts            97.9            2.08
## 11 wicket            90             10
## 12 zookeeper         80             20
```

Mean of percentages of new TDV (mean of repo percentages)

```
## # A tibble: 1 x 2
##   mean_new_tdv_equal_to_zero mean_new_tdv_greater_than_zero
##   <dbl>          <dbl>
## 1      89.6            10.4
```

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`.issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_j$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

```
## # A tibble: 10 x 13
## # Groups:   rank [10]
##   rank accumulo      cayenne 'commons-collections' 'commons-io' 'commons-lang'
##   <int> <chr>      <chr>      <chr>      <chr>      <chr>
## 1 1 java:S1192   ~ java:S... java:S112 - 10 java:S4719 ~ java:S4165 ~...
## 2 2 java:S112   ~ java:S... java:S112 - 6 java:S1192 ~ java:S1192 ~...
## 3 3 java:S1135   ~ java:S... java:S3008 - 5 java:S4042 ~ java:S125 ~...
## 4 4 java:S100    ~ java:S... java:S1874 - 5 java:S899 ~ java:S2293 ~...
## 5 5 java:S2095   ~ java:S... java:S125 - 4 java:S2129 ~ java:S3878 ~...
## 6 6 java:S1155   ~ java:S... java:S1134 - 4 java:S1874 ~ java:S1612 ~...
## 7 7 java:S1659   ~ java:S... java:S1192 - 3 java:S2293 ~ java:S100 ~...
## 8 8 java:S2259   ~ java:S... java:S1126 - 2 java:S1481 ~ java:S1611 ~...
## 9 9 java:S1854   ~ java:S... java:S1905 - 1 java:S125 ~ java:S4968 ~...
## 10 10 java:S3776 ~ <NA> java:S2094 - 1 java:S1854 ~ java:S2129 ~...
## # i 7 more variables: helix <chr>, 'httpcomponents-client' <chr>,
## # 'maven-surefire' <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## # zookeeper <chr>
```

TOP 10 fixed

TOP 10 fixed rules by *PositionScore* metric.

```
## # A tibble: 58 x 2
##   rule      position_score
##   <chr>          <dbl>
## 1 java:S1192          26
## 2 java:S112           70
## 3 java:S1874          74
## 4 java:S100           98
## 5 java:S125          106
## 6 java:S2293         112
## 7 java:S3740         115
## 8 java:S106          116
## 9 java:S1117         119
## 10 java:S1854        119
## # i 48 more rows
```

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

```
## # A tibble: 10 x 13
## # Groups:   rank [10]
##   rank accumulo      cayenne 'commons-collections' 'commons-io' 'commons-lang'
##   <int> <chr>      <chr>      <chr>      <chr>      <chr>
## 1 1 java:S1192   ~ java:S... java:S112 - 295 java:S4719 ~ java:S4165 ~...
## 2 2 java:S112   ~ java:S... java:S125 - 207 java:S100 ~ java:S100 ~...
## 3 3 java:S1135   ~ java:S... java:S108 - 173 java:S899 ~ java:S4165 ~...
## 4 4 java:S1659   ~ java:S... java:S1117 - 109 java:S4842 ~ java:S125 ~...
## 5 5 java:S2095   ~ java:S... java:S2293 - 78 java:S4719 ~ java:S1172 ~...
## 6 6 java:S106    ~ java:S... java:S2281 - 46 java:S2184 ~ java:S3878 ~...
## 7 7 java:S100    ~ java:S... java:S3878 - 45 java:S117 ~ java:S108 ~...
## 8 8 java:S3252   ~ java:S... java:S1854 - 35 java:S100 ~ java:S400 ~...
## 9 9 java:S2184   ~ java:S... java:S112 - 35 java:S3878 ~ java:S1104 ~...
## 10 10 java:S108   ~ java:S... java:S1123 - 33 java:S1874 ~ java:S1186 ~...
## # i 7 more variables: helix <chr>, 'httpcomponents-client' <chr>,
## # 'maven-surefire' <chr>, opennlp <chr>, struts <chr>, wicket <chr>,
## # zookeeper <chr>
```

TOP 10 unfixed

TOP 10 unfixed rules by *PositionScore* metric.

```
## # A tibble: 51 x 2
##   rule      position_score
##   <chr>          <dbl>
## 1 java:S1192          14
## 2 java:S100           86
## 3 java:S106           94
## 4 java:S112          103
## 5 java:S1186         103
## 6 java:S1874         106
## 7 java:S125         109
## 8 java:S2095        110
## 9 java:S108         112
## 10 java:S117        115
## # i 41 more rows
```

Outliers

Higher TDD

```
## # A tibble: 1,152 x 5
##   repo      pr_number total_debt total_ncloc  tdd
##   <chr>      <int>      <int>      <int> <dbl>
## 1 wicket          709        2218        262  8.47
## 2 wicket          708        1161        182  6.38
## 3 struts           807         196         39  5.03
## 4 opennlp          563        1035        232  4.46
## 5 maven-surefire    596        1246        317  3.93
## 6 opennlp          569        1161        325  3.57
## 7 opennlp          585        1296        386  3.36
## 8 commons-lang     1051        1078        324  3.02
## 9 helix           2439         102         34  3
## 10 struts           594         300        101  2.97
## # i 1,142 more rows
```

Higher TDV

```
## # A tibble: 1,152 x 3
##   pr_number repo      tdv
##   <int> <chr>      <int>
## 1 2593 accumulo        2078
## 2 1891 accumulo         643
## 3 483 httpcomponents-client 593
## 4 1605 accumulo         574
## 5 2335 accumulo         557
## 6 559 struts           496
## 7 2680 helix           368
## 8 2289 accumulo         345
## 9 1452 helix           343
## 10 427 struts           333
## # i 1,142 more rows
```

Lower TDV

```
## # A tibble: 1,152 x 3
##   pr_number repo      tdv
##   <int> <chr>      <int>
## 1 484 httpcomponents-client -882
## 2 431 opennlp            -813
## 3 3604 accumulo          -768
## 4 1565 accumulo          -704
## 5 799 struts             -648
## 6 3117 accumulo          -571
## 7 670 struts             -542
## 8 826 struts             -536
## 9 559 opennlp            -528
## 10 1584 helix            -478
## # i 1,142 more rows
```

Extra

Number of PRs with pre-existing TD

```
##   n
## 1 1094
```

Number of java:S1192 issues that affect test code

```
##   n
## 1 18504
```

Number of java:S1192 issues that affect not test code

```
##   n
## 1 84
```

Summary NCLOC

```
##   ncloc
##   Min. :    3
##   1st Qu.: 180
##   Median : 414
##   Mean : 1107
##   3rd Qu.: 1024
##   Max. :61510
```